First Executed Results

The Model-Checked Authority Algebra · the B-21 Audit Benchmark · the Registered Frontier Real-Data Test

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Twelve properties verified, two falsifications honored: Law 2 refuted on one trace; H1 rejected on real data under the programme's own kill criteria. This annex discharges the first standing proof obligations of the SAVANT corpus. Study one executes an explicit-state model check of Paper 1's authority transition algebra, auditing a TOCTOU relaxation alongside the property suite and restating Law 2 in its sound weak form. Study two executes the B-21 seeded-error audit benchmark as a fixed-seed Monte-Carlo experiment. Study three executes the registered real-data test of Blueprint 3's frontier hypothesis against 4,341 CCP amendment events and V-Dem ERT v16.0 labels, with sensitivity 0.00 and specificity 1.00 at the frozen operating point. Every number herein is an output of the shipped scripts; the reproducibility block fixes the code hashes.

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Relation to corpus. Evidence annex to White Papers 1 and 4 and Blueprint 3; discharges the first standing proof obligations and records two honored falsifications (Law 2; H1).

Reproducibility. All numbers are outputs of the shipped fixed-seed stdlib scripts; code SHA-256 hashes in the Reproducibility Block (§A5 / §C5).

How to read this annex

The corpus's claims are issued as proof obligations, not as settled results. This annex discharges the first of those obligations and reports exactly what executed: twelve properties verified across the two studies, and two honest falsifications — Paper 1's Law 2 is refuted on one minimal trace (a specification bug, repaired in its sound weak form), and Blueprint 3's H1 is rejected on real data. In both cases the programme's kill criteria fired and were honored; nothing below has been smoothed, and every number traces to the shipped scripts and their JSON outputs.

Annex Part 1 — Executed Verification Results

SAVANT Constitutional Intelligence Series — Evidence Annex to White Papers 1 and 4

Version 1.0 · first executed results for the corpus · status: all numbers below are outputs of the shipped scripts, reproducible per the Reproducibility Block at the end of this document.

This annex converts two standing proof obligations into executed results:

- Mission 1 — an explicit-state model check of Paper 1's authority transition system (§A1–§A4), discharging or refuting the properties of Paper 1 §4–§7 that are well-defined at a finite abstraction.
- Mission 2 — the B-21 seeded-error audit benchmark of Paper 4, executed as a Monte-Carlo experiment with forced depths 1–5, correlated auditor failure, declared costs, and bootstrap confidence intervals (§B1–§B4).

Both scripts are pure Python 3 stdlib, fixed-seed, and self-contained. Where a property fails, the counterexample is printed verbatim and the failure is diagnosed as a spec bug or an abstraction artifact. Nothing below is smoothed.

§A1 Abstraction specification

Paper 1 §2.4 asserts that the machine "must be finite" for verification. As stated, that claim is false for the unabstracted system : the state record $\sigma = \langle \varphi, A, \Xi, Q, H \rangle$ contains the lineage chain H, which grows on every operator application (AX9: audit precedes effect, always), and the pending-event queue Q is likewise unbounded. A transition system with an append-only history component has infinitely many reachable states. The finiteness claim survives only after projection. This annex is that fix : every continuous or unbounded component is discretized or projected, the projection is stated here in full, and each property verdict carries its dependence on the abstraction.

State vector `s = (phase, holder, a, E, U, L, vH, clock, dissent, confirmed, cls, adverse, pend)`:

Component	Domain	Abstracts	Soundness note		
phase	9 lifecycle phases + Terminal	Paper 0 §5.2 control location	exact; Acquire folded into GRANT (no authority-relevant distinction in Table A-1)		
holder	{none, machine, human}	single-holder invariant I-SH	exact at atomicity assumption		
a	{0..4}	grade a∈[0,1]	monotone direction preserved; decision boundaries refined		
E	{0..4}	decayed evidence quality	as above		
U	{0,1}	uncertainty rise/fall deltas	booleanized		
L	{0,1}	legitimacy chain intact (AX5)	booleanized		
vH	{0,1}	lineage integrity verified(H) (I7)	projection: lineage length projected away; no rule in Paper 1 §2–§3 conditions authority on \	H\	, only on verified(H), so this projection is exact for every property checked
clock	{0,1,2}	expiry epochs (τ mandatory finite, AX8)	deadline epochs per §2.4		
dissent, confirmed	{0,1}	queue content relevant to SUSPEND / T-3 gate	queue projected to its authority-relevant bits		
cls	{C1, C2}	consequence class	two classes suffice to exercise the C2+ gate		
adverse	{0,1}	monitor, not machine state: 1 iff an adverse delta occurred with no human-family event since	exists only to make Law M a checkable transition property		
pend	{0..4}	pending grant payload; identically 0 in the atomic model	used only by the TOCTOU variant (§A4)		

Evidence–authority coupling. `bound(E,U) = max(0, E−U)`. GRANT sets `a = min(requested, bound)`; adverse deltas set `a' = min(a, bound)` immediately and unprivileged (rule [CONTRACT], AX4); `e_up`/`u_down` never raise `a` (§7.2 hysteresis: re-expansion only via fresh human GRANT). The γ≥1 coupling exponent is realized as this step bound; what the checked properties require is the monotone direction, which this realizes exactly.

Why this direction is sound for safety properties. Every safety property checked is monotone in the abstracted quantities (Paper 1 §2.4's own argument): widening the lattice cannot introduce a forbidden state that the lattice lacks, provided every guard decision boundary is representable — and all guards here use only `bound(E,U)`, `L`, `vH`, `clock`, and phase, all of which are exact at this abstraction. No soundness claim is made for properties outside the checked set (T-0 remains open in both papers' sense; see §B4).

Enumeration. Exhaustive BFS from σ_0 = (Constrain, none, a=0, E=2, U=0, L=1, vH=1, clock=0, …). Atomic model: 3,062 reachable states, 18,532 transitions , 0.07 s. Transition labels span all seven operators plus environment deltas, human actions, and internal events per Table A-1.

§A2 Property table (atomic model)

Property	Statement (condensed)	Verdict	Evidence
T-1 (Law M)	No component of A increases between an adverse delta and the next human event	VERIFIED	18,532 transitions examined; zero with adverse=1 ∧ non-human ∧ a′>a
LAW-M+ (strong form)	Every grade-increasing transition is human-gated	VERIFIED	subsumes no-resurrection-after-EXPIRE-without-GRANT; zero violations
LAW-1	U↑ never increases a	VERIFIED	all u_up transitions: a′≤a
LAW-2 (strict)	∀ E↓: a′<a ∨ a′=a=0	FALSIFIED	counterexample below, §A3 — spec bug, not abstraction artifact
LAW-2-weak	∀ E↓: a′≤a	VERIFIED	the sound restatement (O-1 monotonicity)
LAW-3	L↓ chain break ⇒ a′=0	VERIFIED	all l_down transitions zero the grade
I-SH	Exactly one active holder at every instant	VERIFIED-BY-CONSTRUCTION	holder is a single-valued field; TRANSFER is one atomic edge. Discharges I-SH only up to atomicity — see §B4 and §A4
T-4	¬verified(H) ⇒ a=0	VERIFIED	3,062 reachable states, zero with vH=0 ∧ a>0
T-3	C2 execution requires human confirmation	VERIFIED	zero execute transitions from cls=C2 with confirmed=0
T-6	Adverse delta from Exercise/Verify reaches Constrain in one step	VERIFIED	zero interposable transitions
T-9	Recovery restores authority only on proven chain; RECOVER itself yields none	VERIFIED	no recover/countersign transition sets a>0; no reachable Recover/Reauthorize state carries a>0; countersign guard requires vH=1
CU	From every reachable machine-authority state a contraction-family transition is enabled	VERIFIED	all machine-holding states examined; CONSTRAIN's guard is true (AX4)
T-5	Termination of authority: from every machine-authority state a non-human path to a=0 exists	VERIFIED	reverse BFS over env/op edges only; clock fairness + mandatory τ suffice (O-3)
T-7	Escalation reaches Adjudicate or fail-safe, never limbo	VERIFIED	escalate_timeout enabled in every Escalate state; conditioned on timeout fairness only, per §2.5
T-10*	Lattice form: every machine-holding state has a ≤ bound(E,U)	VERIFIED	3,062 states, zero violations. The timed lag δ of full T-10 is not modeled (§B4)

Not checked at this abstraction (stated, not hidden): T-0 (meta-soundness), T-2 (execute-premises in full — partially covered by T-3 and the [EXEC] guard), T-8 in its queue-priority form (the queue is projected to bits; dissent→Escalate holds in one step in the model but starvation-freedom under queue load is out of scope), T-10's timed lag, Laws 4–5 (capability/invariant vectors and multiplicative composition are not in the state vector).

§A3 Counterexamples and findings

Finding 1 — Law 2's strict form is false (real spec bug).

```
phase=Exercise holder=machine a=1 E=2 U=0 ... →[e_down]→ phase=Constrain a=1 E=1 ...
```

Evidence quality fell (E: 2→1); grade stayed at 1, nonzero. The paper's own kill condition is met verbatim: "a trace where evidence quality falls between consecutive states with grade unchanged and nonzero." Diagnosis: the contraction operator `a′ = min(a, bound(E,U))` is monotone (O-1), not strictly decreasing. Strictness holds only where the evidence term binds — i.e., where a equals the pre-delta ceiling. Whenever grade sits strictly below the ceiling (the normal case after any prior contraction), a one-tier evidence drop leaves the grade unchanged. The continuous form `a = g_E(E)·G(...)` has the same structure: ∂a/∂E > 0 only where g_E binds. Law 2 survives only in its weak monotone form (verified above); the strict form should be restated as "a′ ≤ a, with strict inequality whenever a attains the evidence ceiling." This is a one-trace refutation of the kind Paper 1 §12.2 (F-d2) says must change the theory.

Finding 2 — modeling correction during this work (recorded honestly). An early version of this model let `adjudicate_reauthorize` from Escalate preserve machine authority, producing a reachable Reauthorize state with a>0 — a T-9 violation. Trace inspection showed the model, not the paper, was wrong: §6.3 and Trace 2 unambiguously strip machine authority at adjudication handoff (withdrawal = empty permitted set). The transition was corrected (adjudication moves holder to human/none, a→0; re-entry only via fresh GRANT), after which T-9 verifies. The episode is reported because it demonstrates the checker's sensitivity: a single mis-wired edge produced a reachable violation of exactly the property Paper 1 calls "the dangerous direction."

§A4 The TOCTOU variant (check/effect gap)

Paper 1's O-2 and benchmark B-09 flag the seam between gate evaluation and effect execution as an empirical-only vulnerability: a grant evaluated against fresh context but applied after adverse deltas land would resurrect stale authority. This annex makes that concrete. The perturbed model splits GRANT into `authorize_check` (human; evaluates the guard against current context, records `pend = min(req, bound)`) and `apply_grant` (internal; applies the recorded grant later, without re-check), with environment deltas free to interpose between them.

Result (5,805 states, 38,863 transitions): four properties that hold in the atomic model fail , each with a short counterexample trace:

Property	Atomic	TOCTOU	Interposing event in the trace
T-1 (Law M)	VERIFIED	FALSIFIED	e_down between check and apply; apply_grant raises a inside the adversity episode
LAW-M+	VERIFIED	FALSIFIED	apply_grant is an internal event that increases a
T-4	VERIFIED	FALSIFIED	lineage_corrupt between check and apply; grant applied with vH=0
T-10*	VERIFIED	FALSIFIED	e_down between check and apply; a=2 executed with bound(E,U)=1

Shortest T-1 counterexample (verbatim from the checker):

```
[authorize_check(req=1)]  phase=Constrain holder=none a=0 E=2 ... pend=0
    -> phase=Constrain holder=none a=0 E=2 ... pend=1
[e_down]                  -> phase=Constrain ... E=1 ... adverse=1 pend=1
[apply_grant]             -> phase=Exercise holder=machine a=1 E=1 ... adverse=1
```

Interpretation: the contraction laws are not robust to asynchronous gate evaluation . Every safety guarantee of §4–§5 that this annex verified depends on check-and-effect atomicity. In any implementation, grant application must re-validate against current context or hold a lock across check and effect — this converts Paper 1's B-09 from a prose warning into a demonstrated four-property collapse, and identifies precisely which properties carry the risk.

§B1 B-21 protocol

Synthetic-but-principled Monte-Carlo of the Paper 4 audit stack, forced depths 1–5 (the §7.4 lab arm; the stopping predicate is not applied). All parameters are declared modeling choices, not measurements; the full set is echoed in `results-b21.json`.

- Ground truth: R1 errs with probability 0.20; seeded errors typed — contradiction (40%), stale-evidence (35%), scope-overreach (25%).
- R2 sensitivity per type: 0.85 / 0.80 / 0.70 (base regime) and halved (hard regime, testing where depth 3 pays). False-amendment rate on clean decisions 0.04; a false amendment corrupts with probability 0.5.
- Rk, k≥3: intercepts a surviving genuine error with probability `SENS[k]·(1−ρ)`, SENS = 0.75/0.70/0.65 for k=3/4/5; ρ is the shared-failure correlation (probability that the previous level's miss has a cause shared with level k — §7.2, I-DI). Reverts a predecessor's false amendment with probability REVERT[k] = 0.80/0.75/0.70 (not ρ-discounted). False-amendment rate FP[k] = 0.04/0.05/0.07/0.10, non-decreasing — Eq. 7-1 clause 2 as a parameter.
- Cost: per-level 1.0 / 1.5 / 2.5 / 4.0 / 6.0 audit-cost units. Harm H=10 per released wrong decision. `V(d) = −10·err(d) − λ·Σcost`; `ΔV(k) = V(k+1) − V(k)` (Eq. 6-2).
- Sweep: ρ ∈ {0.0, 0.2, 0.5, 0.8} × λ ∈ {0.05, 0.2, 1.0} × 2 sensitivity regimes × depths 1–5. N=2000 trials per (regime, ρ); depths paired within trial; bootstrap 95% CIs, B=1000 genuine resamples with replacement (joint over depths and statistics), percentile method. Seed 20261004.

§B2 Results

Released-error rate err(d) by forced depth (bootstrap CIs in `results-b21.json`; base regime):

ρ	d=1	d=2	d=3	d=4	d=5
0.0	0.198	0.052	0.043	0.037	0.060
0.2	0.203	0.063	0.045	0.043	0.053
0.5	0.193	0.060	0.050	0.054	0.068
0.8	0.192	0.053	0.052	0.067	0.079

Hard regime (R2 sensitivity halved):

ρ	d=1	d=2	d=3	d=4	d=5
0.0	0.193	0.126	0.055	0.045	0.062
0.2	0.205	0.138	0.077	0.052	0.059

0.5	0.199	0.135	0.100	0.083	0.088
0.8	0.195	0.138	0.126	0.127	0.129

Note the non-monotone tail: err(d=5) exceeds err(d=4) in six of eight regimes — the self-referential-error clause made visible; depth 5 damages more than it catches.

Correctness gains g_k = err(k)−err(k+1) (Eq. 7-1 clause 1; CI95):

regime	ρ	g_1 (1→2)	g_2 (2→3)	g_3 (3→4)	g_4 (4→5)
base	0.0	0.146 [0.131, 0.163]	0.009 [−0.003, 0.021]	0.006 [−0.004, 0.018]	−0.023 [−0.035, −0.013]
base	0.8	0.139 [0.121, 0.156]	0.001 [−0.007, 0.009]	−0.015 [−0.025, −0.004]	−0.012 [−0.024, 0.001]
hard	0.0	0.067 [0.055, 0.079]	0.071 [0.057, 0.086]	0.010 [−0.001, 0.021]	−0.017 [−0.029, −0.005]
hard	0.2	0.068 [0.054, 0.081]	0.060 [0.046, 0.074]	0.026 [0.013, 0.037]	−0.007 [−0.019, 0.005]
hard	0.5	0.065 [0.051, 0.077]	0.035 [0.023, 0.047]	0.017 [0.004, 0.029]	−0.006 [−0.018, 0.007]
hard	0.8	0.057 [0.044, 0.070]	0.012 [0.003, 0.022]	−0.001 [−0.011, 0.011]	−0.003 [−0.015, 0.011]

Gains are non-increasing in k in all eight regimes (H30 clause 1 holds in this model — it was built in via declining SENS and rising FP; see §B4 for what this does and does not show).

Marginal value ΔV(2) = V(3)−V(2) and optimal depth (all 24 cells; bold = CI excludes 0):

regime	ρ	λ=0.05: ΔV(2), opt	λ=0.2: ΔV(2), opt	λ=1.0: ΔV(2), opt
base	0.0	−0.035 [−0.15, 0.08], d2	−0.410, d2	−2.410, d1
base	0.2	+0.055 [−0.055, 0.17], d3*	−0.320, d2	−2.320, d1
base	0.5	−0.020 [−0.12, 0.075], d2	−0.395, d2	−2.395, d1
base	0.8	−0.115, d2	−0.490, d2	−2.490, d1
hard	0.0	+0.585 [0.44, 0.735], d3	+0.210 [0.065, 0.36], d3	−1.790, d1
hard	0.2	+0.475 [0.34, 0.62], d4*	+0.100 [−0.035, 0.245], d3*	−1.900, d1
hard	0.5	+0.225 [0.105, 0.345], d3	−0.150, d2	−2.150, d1
hard	0.8	−0.005 [−0.10, 0.095], d2	−0.380, d2	−2.380, d1

(\ point estimate only; CI includes 0. In the one cell where depth 4 is point-optimal — hard, ρ=0.2, λ=0.05 — ΔV(3) = +0.055 with CI [−0.07, 0.175]: not significant. Depth 4 significantly beats depth 3 in zero of 24 cells *.)

Interception. Base regime cumulative interception of seeded errors: 78% at depth 2, 90% at depth 3, 97% at depth 4 (ρ=0); at ρ=0.8 the depth-3 marginal interception collapses (83% cumulative at depth 3 vs 80% at depth 2). False-amendment rate on clean decisions rises from ~0.03 at level 2 to ~0.08 at level 5, all regimes — parameter-driven, reported, not claimed as evidence for H31.

§B3 Verdict on R_max = 3

Conditional, in both directions, exactly as the data says:

1. Where depth 3 beats depth 2: only in the hard-residue regime (weak R2) with low auditor correlation and cheap audit: ρ ≤ 0.5 and λ ≤ 0.05–0.2. There ΔV(2) > 0 with CI excluding 0 (up to +0.585 harm units). This matches the paper's own statement of when the third level earns its latency: "the cases where the audit of the audit catches what the audit missed" — but the data adds a precondition the prose underweights: the second level must have missed enough to matter.

2. Where it fails: (a) base regime — a competent R2 (sensitivity ≈ 0.8) leaves residue too thin for depth 3 to pay at any λ; depth 2 is optimal or statistically tied in all 12 base cells. Per §1.2 of the paper, the honest response in such a domain is to tighten the bound to 2, and this data says exactly that. (b) ρ ≥ 0.8 — correlated auditors destroy the third level's value regardless of cost; repetition does not help when the miss is shared-caused. (c) λ = 1.0 — when audit cost prices anywhere near error harm, depth 1 is optimal in all 8 high-λ cells; audit is a luxury only cheap time affords (§7.3 made quantitative).

3. As a hard bound against depth ≥ 4: supported here. Depth 4 significantly beats depth 3 in 0 of 24 cells, and depth 5 is significantly worse than depth 4 in six of eight regimes. R_max=3 as a ceiling survives; R_max=3 as a universal optimum does not.

Net: the paper's treatment of R_max=3 as an amendable experimental hypothesis rather than dogma is the correct posture, and this benchmark demonstrates both branches of the amendment logic — tighten where residue is thin, hold the ceiling where depth-4 value never reaches significance.

§B4 Limitations

1. Abstraction dependence (Mission 1). Verdicts are discharged for the lattice model, not the continuous refinement; T-0 (soundness of the checking model) remains an open obligation. Monotone safety properties transfer upward by the paper's own §2.4 argument; no such claim is made for T-8's starvation form, T-10's timed lag δ, Laws 4–5, or H9–H14 (empirical, out of model-checker scope). I-SH holds by construction; the concurrent-transfer storm (B-10) needs a non-atomic-transfer model, which this annex does not build.

2. Queue/lineage projection. H is an integrity bit and Q is two bits. Properties that could depend on queue ordering or lineage content are unchecked by construction.

3. Simulation, not measurement (Mission 2). B-21 as executed here is a synthetic corpus with declared error frequencies and sensitivities. It demonstrates the decision logic of the depth question and the regime structure of the answer; it does not measure any real audit stack. The fact that Eq. 7-1's shape appears in the output is substantially because it was built into the parameters — the non-trivial content is where the value crossover lands under that shape, which is a computed result, not an assumption. Claiming H29–H31 confirmed from this run would be overreach and is not done.

4. λ and H are unit conventions. The regime boundaries move with H/λ ratios; the qualitative structure (depth 3 pays iff residue × catchability exceeds priced cost) is what transfers.

5. TOCTOU variant models one asynchrony. Other interleavings (concurrent TRANSFER/EXPIRE races, crash mid-audit-append) are flagged by Paper 1 (B-10, H11–H12) and remain unexecuted obligations.

Reproducibility block

Item	Value
Python	3.12.12
modelcheck.py SHA-256	35dbebe3e6017f4ae05d46e9c31ffec7a763c4c255e4977cc5d8c2acd48bc76f
b21.py SHA-256	8bd9baac7da45bdecca11b9913fa1b8e17f673382fd5022e7742b6420ab944aa

Runs	python3 modelcheck.py → results-modelcheck.json (3,062 states / 18,532 transitions, 0.07 s); python3 modelcheck.py --toctou → results-modelcheck-toctou.json (5,805 / 38,863, 0.15 s); python3 b21.py → results-b21.json (16,000 trials total: 2 regimes × 4 ρ × 2,000, each trial forced through depths 1–5; plus 8,000 bootstrap replicates of 2,000 resamples each, ≈13 s)
Seeds	model check: deterministic (no randomness); B-21: base seed 20261004, per-regime/ρ derived, bootstrap seeds derived; all derivation in code
Code hashes	embedded in each results JSON (code_sha256 field) and verified to match sha256sum of the shipped files at write time

Annex, Part 2 — The CCP Real-Data Run, Executed

SAVANT Blueprint Nº 3, §6.3 — external empirical terminus. Ladder level reached: L2 (ρ-only variant on real CCP-derived telemetry, plus the λ-unavailability finding).
Headline verdict: H1 REJECTED at the frozen operating point (k = 1.0, h = 4.0, d = 2); the registered family is likewise rejected on real data (0 of 72 registered points feasible). Kill criteria K1 and K2 both trigger, and are reported as written.
This annex reports exactly what the data said. No numbers below are adjusted, selected, or rounded in the programme's favor; the headline is a failure, and it is stated as one.

1. What was executed

The frozen instrument of Blueprint Nº 3 §4.1/§6.1 — per-system baseline of the first 8 quarters, one-sided CUSUM with drift allowance k = 1.0, alarm threshold h = 4.0, persistence d = 2 consecutive quarters, baseline quarters ineligible for alarm — was carried unchanged to real data. Telemetry ρ (version-flip frequency) was constructed from the CCP-derived chronology of constitutional events; episode labels were taken from the V-Dem Episodes of Regime Transformation (ERT) dataset v16.0. The frozen criteria (Blueprint Table 2) were applied unchanged:

Criterion	Frozen threshold	Role
C1	Association $p < 0.05$ against a 10,000-permutation null	signal exists
C2	Specificity ≥ 0.85 on control systems	false-alarm discipline
C3	Sensitivity ≥ 0.70 on capture systems	detector must detect
C4	≥ 50% of detections alarm ≥ 4 quarters before failure	must lead
C5	Control alarms must not cluster in the baseline window	hygiene (qualitative)

Verdict rule (frozen): SUPPORTED iff C1–C4 all pass; PARTIALLY SUPPORTED iff C1–C3 pass and C4 fails; REJECTED otherwise.

2. Data provenance

Full detail, checksums, and the access log (including every failed source) are in `data/SOURCES.md`. Summary:

- Episodes (labels): V-Dem ERT v16.0, `inst/ert.csv` from github.com/vdeminstitute/ERT, SHA-256 `26e805df…`, retrieved 2026-10-12. Country-year 1900–2025. Autocratization episodes and outcomes per the ERT codebook; capture label = `aut_ep_outcome == 1` (democratic breakdown); failure year F = first episode year with `v2x_regime ≤ 1` (RoW electoral or closed autocracy).
- Telemetry (ρ): Constitute Project chronology dump (`constitute_chronology.xml`, SHA-256 `65f7deba…`, retrieved 2026-10-12), the publication venue of the Comparative Constitutions Project's chronology of constitutional events. 4,341 events, 223 countries, through 2026. Version-flip events = `New` (replacement) + `Amendment`. The canonical CCP download is registration-gated and returned HTTP 403 from this environment; the BERD@NFDI mirror carries only 100-line previews. The Constitute dump is the same CCP chronology, openly licensed, and is used in its place — noted as a provenance substitution, not concealed.
- **λ (cooling-off compliance): not obtained, because it does not exist in any public source.** See §5.

3. Corpus

- Capture systems (n = 42): uncensored ERT democratic-breakdown episodes with failure year F ∈ [1990, 2024], the country present in the chronology, and complete ERT coverage of the 15-year window [F−14, F]. 29 breakdown episodes were excluded by these rules (25 with F < 1990 — the Weimar, 1930s European, and 1960s–80s classes — and 4 with window coverage gaps: Armenia 1995, Belarus 1996, Estonia 1992, Ukraine 1998); all exclusions are listed in `results-ccp.json`.
- Control systems (n = 101): countries with complete ERT rows for 2005–2019 and no autocratization episode overlapping that window. Control window fixed at 2005–2019 (60 quarters).
- Averted-episode class (n = 12): episodes ending in preempted breakdown or averted regression (`aut_ep_outcome ∈ {2,4}`), reported separately and not scored in C2/C3 (e.g., Poland 2015–2023 belongs here).
- Corpus membership and failure years were sealed to `data/labels_sealed.json` (SHA-256 `3b57231a…`) before any scoring. Blinding is simulated — constructor and scorer are the same analyst (deviation D4).

4. Deviations from the frozen protocol

All fixed before scoring; none reversible after.

- D1 — Temporal resolution. CCP/Constitute events carry year resolution only. Quarterly ρ was constructed as the honest annual rate (events-in-year / 4, constant within year); no intra-year timing was fabricated. Window geometry (60 quarters, first-8-quarters baseline) is preserved. A yearly-step construction (baseline = first 2 years = 8 quarters, leads ×4) was run as robustness variant R1.
- D2 — ρ-only variant. The conjoint detector requires λ, which is unavailable (§5). The alarm rule used is the ρ-channel conjunct alone: `ALARM(t) ≡ S_ρ(τ) > h for all τ ∈ {t−d+1,…,t}`. This is the pre-authorized L2 fallback, labeled as a deviation.
- D3 — Degenerate baseline. The frozen protocol does not specify σ_base = 0 handling. Chosen before scoring: z_t = ρ_t − μ_base (unstandardized deviation). 98 of 143 scored system-windows hit this fallback — itself a real-data finding (see §7).
- D4 — Simulated blinding (above).
- D5 — Window geometry. Capture windows end at the failure year, so the negative-lead hygiene rule is vacuous by construction.

5. The λ-unavailability finding (a result, not an excuse)

Cooling-off compliance λ — the degree to which mandated deliberation intervals are actually observed — is not coded by any public dataset . CCP Characteristics codes de jure amendment procedure (proposal/approval rules, `amndprop_ `, `amndappr_ `, `amndamaj`, `amndapct`; verified in the BERD preview), not de facto observance of delays. Constitute codes nothing of the kind. No public source records whether a constitutionally mandated waiting period between amendment readings was actually honored.

Consequence for the programme, stated plainly: **the conjoint instrument's second channel requires primary data collection that does not yet exist.** The blueprint's §6.3 estimate of "days of acquisition and blind coding" is, on the evidence of this attempt, optimistic for λ: the telemetry must be created, not downloaded. This is the construct-validity risk the programme's critics flagged, now documented on real data, and it is a finding this annex claims as part of the result.

6. Results at the frozen point

Primary construction (quarterly rate, D1), frozen point (1.0, 4.0, 2), seed 20260927, 10,000 permutations:

Criterion	Frozen threshold	Result	Verdict
C1 association	$p < 0.05$	$p = 1.000$ (TP = 0; null mean 0.0, null max 0)	FAIL
C2 specificity	≥ 0.85	1.000 (101 TN / 101 controls; 0 false alarms)	PASS
C3 sensitivity	≥ 0.70	0.000 (0/42 capture systems detected)	FAIL
C4 lead	≥ 50% of detections at ≥ 4q	no detections; undefined, scored 0	FAIL
C5 hygiene	no baseline clustering	vacuous — no alarms of any kind	vacuous

Verdict: H1 REJECTED at the frozen point, unchanged, on real data. Robustness variant R1 (yearly construction) is identical: TP = 0, FP = 0, REJECTED. Robustness variant R2 (counting `Interim` constitutions as version-flips) is identical: TP = 0, FP = 0. Averted-episode class: 0 of 12 alarmed.

Under the programme's own pre-registered kill criteria: K1 triggers — the feasibility result of A2 is corpus-bound, and H1 returns to REJECTED at this standard. Reported as such, per the A2 epistemic contract (Blueprint Appendix D).

7. Why the detector was silent — a structural diagnostic, not a near miss

The failure is not marginal. Across all 155 scored and reported system-windows (42 capture, 101 control, 12 averted) and all quarters, the standardized ρ increment never exceeds the drift allowance:

- max $z_\rho = 1.00$ over every system-window and every quarter, in both constructions. With $k = 1.0$, the CUSUM increment $z - k \leq 0$ everywhere; S_ρ is identically zero for every system. max $S_\rho = 0.00$.
- Root cause: telemetry quantization. The CCP chronology codes at most one version-flip event per country-year (verified: the raw file's maximum is 1). ρ is therefore a sparse {0, 1}/year process. Against any own-baseline, a single-event year yields $z \leq 1.0$; accumulation to $h = 4.0$ would require multi-event years the coding structurally cannot represent.
- The synthetic corpus's ρ channel carried continuous version-flip rates with capture-phase drift $z \approx 1.5$–1.6 per quarter (Blueprint Appendix E). That dynamic range does not exist in the CCP telemetry the corpus was anchored to.

* Corpus R reproduced the shape of capture histories but not the scale

properties of the real telemetry source. * This is a corpus-construction finding, and it is the substantive content of the K1 verdict.

8. K2 — the registered family on real data

The A2-registered grid (k ∈ {0.25, 0.50, 0.75, 1.00} × h ∈ {3,4,5,6} × d ∈ {1,2,3}; 72 points) was evaluated under the same sealed labels, as the pre-registered K2 test requires ("no registered operating point achieves C1–C4 simultaneously on real data"):

- Feasible points: 0 of 72 (quarterly construction; 0 of 72 yearly).
- Maximum sensitivity at any registered point: 0.19 (8/42, at k = 0.25, h ≤ 5.0), against the frozen C3 floor of 0.70. Specificity at those points: 0.85.
- Permutation nulls on real data are honest and behave correctly: at the most sensitive point (k = 0.25, h = 3.0, d = 1), observed TP = 8 against a null mean of 6.74 (p = 0.344); the synthetic-run null means (6.62–9.03 against TP = 18) do not recur because there is no real-data signal to find. No registered point attains p < 0.05 except none — minimum observed p = 0.0503 (k = 0.5, h = 5.0), marginal and at sensitivity 0.12.
- Full per-point results: `results-ccp.json`, keys `K2_grid_quarterly`, `K2_grid_yearly`.

K2 triggers: on this telemetry, the Ω-family is rejected as a leading indicator. Per the registered wording: the framework retains its theorems and loses this instrument — reported as such.

9. What did replicate

Two partial replications deserve honest mention alongside the failure:

1. The German-class watch finding replicates. At the most sensitive registered points (k = 0.25), the 15 control false positives are dominated by high-cadence reform democracies — Germany, Austria, Belgium, France, Norway, Chile, Colombia — exactly the class the A2 report's specificity ceiling named. The instrument's known false-positive anatomy is real.
2. Specificity direction. Real-data specificity at the frozen point is 1.00, above the corpus ceiling of 0.90 — because the detector never fires at all. A specificity "pass" bought with zero sensitivity is not a capability; it is reported as the trivial endpoint it is.

The eight capture systems detected at the most sensitive point (Burkina Faso 2015 & 2022, India 2017, Moldova 2005, Malawi 2000, Nicaragua 2007, Papua New

Guinea 2008, Türkiye 2013) are plausibly the class with visible formal amendment cadence — but at sensitivity 0.19 and p = 0.34 this is description, not detection.

10. Limitations

1. ρ-only detector (D2): the conjoint instrument was never executable on public data; this run bounds the ρ channel only.
2. Year-resolution events (D1); quarterly construction distributes events uniformly within year. z-scores are scale-invariant to this transform, but the D3 fallback is not (quarterly fallback z = 0.25 vs yearly z = 1.0 for empty baselines) — both variants are reported and agree.
3. Exclusion of all pre-1990 breakdowns (Weimar, 1930s Europe, Chile 1973, etc.) removes the historical classes from which the corpus was parameterized; amendment coding quality pre-1990 motivated the cut, but the cut is a choice, made before scoring.
4. ERT v16 codings of very recent breakdowns (e.g., Mexico, Indonesia, Mongolia, Georgia, all F = 2024) are contested in the literature; labels were taken as published, per frozen-label discipline.
5. Control windows are fixed at 2005–2019 while capture windows vary (1976–2024); epoch effects on amendment cadence are not controlled.
6. Blinding was simulated (single analyst). The seal fixes labels before scoring but cannot substitute for an independent coder.
7. Constitute's chronology is CCP-derived but may lag the canonical CCP release; the 403-gated canonical file could not be checksum-verified against it.

11. Requirements for the L1 run (the honest shopping list)

1. A λ telemetry source: primary coding of actual proposal-to-promulgation intervals against constitutionally mandated cooling-off periods, per amendment event, per country. This is a data-collection project (weeks to months of coded labor), not a download.
2. Sub-annual event timing (promulgation dates) to restore quarterly ρ without interpolation — available in principle from constitutional texts, not coded in CCP.
3. A registered amendment re-specifying the ρ channel's units: any detector whose drift allowance assumes z-drift ≈ 1.5/quarter cannot run on {0,1}/year telemetry. Under the refusal set, that amendment must precede any re-run.

12. Reproducibility

```
code:      ccp_run.py            SHA-256 cc4b242ec8098ec3252bdf787b596fdb990d869ebf112a8b34e6eb0312758bfc
labels:    data/labels_sealed.json   SHA-256 3b57231ab6fa828a44e12efc6c79839123d857b51a5ea30cf114be371d83aa99
episodes:  data/ert.csv          SHA-256 26e805df073a26973f5c0fbefdcdd0d20776245d71092a575063e47ffac6db5f
telemetry: data/constitute_chronology.xml  SHA-256 65f7debafff9a36a08dd3058807c94f904bdee963493c2affc8fd9a7d2a2f98
seed:      20260927   permutations: 10,000   runtime: ~4 s (Python 3.12, stdlib only)
command:   cd annex && python3 ccp_run.py   ->  results-ccp.json (deterministic, verified byte-identical across ru
output:    results-ccp.json      SHA-256 4eee02d13ef2fa80b1cafd76e14c1fb947ac17e7ecf49475d6573c3333003e51
```

13. Closing note

The blueprint's Part VII pre-wrote this announcement: if the frozen point fails C1–C4 unchanged on real CCP data, the feasibility result is corpus-bound and H1 returns to REJECTED — reported as such. It failed, and it is so reported. The run additionally produced two findings the corpus could not: (i) the conjoint channel's λ telemetry does not exist publicly and must be created; (ii) the ρ channel's registered operating point is structurally inert on CCP annual telemetry (max z = 1.0 = k; S ≡ 0), a scale mismatch between the synthetic corpus and the real source it was anchored to. The German-class false-positive anatomy replicates. Under the programme's own kill criteria this is a clean, executed falsification — the instrument is lost, the discipline held, and the record states both.